

Functional training methods for the runner's myofascial systems

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Introduction

Running is big business and preparation is paramount to success in the sport. Participation has increased exponentially in the last 3 decades at both the amateur and the professional level. On average, runners cover ~110–130 km a week but, at times, can be thwarted by a variety of injuries. First, it should be mentioned that running is a complex elastic/spring-like movement involving the whole body's gait mechanism. An improved efficiency or energy-saving motion can be achieved when close to ideal elastic bounce in the running or walking gait is attained (Chapter 8). The beauty of functional myofascial training is that it can not only prevent injuries but can also increase running efficiency and myofascial tissue fatigue resistance.

This chapter focuses on how functional myofascial training can address the elastic strength and coordination components of the running and walking mechanism (Chapter 30). Reference will also be made to the principles of biotensegrity (Chapters 11 and 36) as a means of explaining the mechanism of running and the behavior of the body tissues, which allow it to happen. Briefly, biotensegrity explains how the principles of tensegrity (Chapter 11) manifest in biological systems: from viruses to cells and tissues of living systems (i.e. plants and animals). Biotensegrity integrates complex anatomy and biomechanics to make sense of living systems as a functional unit. From a biotensegrity perspective, any body movement (including walking, running and sprinting) is a continuous balancing dance of

tension and compression forces within the body systems (Levin, 2002; Scarr, 2014).

Running injuries are related to poor running technique and coordination, minimal or poor elastic bounce, muscle (myofascial complexes) weakness (e.g. hip abductors, quadriceps/knee mechanism), myofascial complexes imbalances in running structure (e.g. pelvic trunk myofascial complexes), biomechanical faults (over-pronation of feet, valgus of the knee), micro-trauma from overuse, an inadequately trained elastic myofascial net, low resistance to tissue fatigue and overall diminished body global strength relative to impact cyclic loading. All of the above can influence each other, creating a collage of epidemiological causes for running injuries. Studies show that the majority of running injuries can be summed up as micro-trauma to collagenous tissues (Stanish, 1984; Elliott, 1990).

It has been well documented that over 70% of recreational runners will sustain an injury during a 1-year period (Caspersen et al., 1984; Rochcongar et al., 1995; Ferber et al., 2009). For instance, more than eight out of 10 running injuries are below the knee, suggesting some common mechanism might be the culprit (Ferber et al., 2009). Based on the biotensegrity model, evidence does not support any one segmental region but more a global involvement of myofascial-skeletal running structures. Excessive pronation as a causative factor in overuse injuries is well documented (Clarke et al., 1983; Messier et al., 1991; Ferber et al., 2010.). Also, hip and pelvic complex mecha-

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nism weakness and imbalance, or poor stabilization, are now believed to be one of the major links to lower body running injuries: for example, iliotibial band compression syndrome and patella femoral syndrome (Horton and Hall, 1989; Livingston, 1998; Fredericson et al., 2000; Mizuno et al., 2001; Witvrouw et al., 2000). The aforementioned studies indicate that the causes of the majority of running injuries are related, or have some link to, inefficiency in the structural integrity of the running mechanism (the myofascialskeletal system). It is, therefore, clear that the prevention, pre-habilitation and rehabilitation of such injuries must address these causes in a practical manner. This leads to the proposed functional approach to train the runner's fascia from a global prospective.

However, at this point, the following questions must be addressed:

- What are the principles of running and how do they relate to global functional training?
- What do we mean by functional training?
- What is the purpose of strength training for runners?
- What do we mean by myofascial training?
- What functional myofascial training is specific to runners?

What are the principles of running and how do they relate to global functional training?

Walking, running and sprinting are complex elastic movements involving the entire body. They involve a cyclic exchange between potential and kinetic energy (a storage and release mechanism). The bones, tendons and ligaments are some of the stiffest springs in the body. The movement in such activities is not incremental or segmental but involves the entire body in

simultaneous global action. Hence, training runners and sprinters cannot merely include exercises, which target segmental body parts (e.g. muscle groups- hamstrings, calves or core-meaning abdominal muscle strength; i.e. six-pack look). Evidence has shown that such exercises do not target the true bio-movements of running and sprinting in an effective and functional manner. In fact, this type of segmental training creates more myofascial and biotensegral imbalances setting up a platform/environment in the body's structure/architecture making it more prone to injury and substandard performance (Huijing et al., 2007; Stecco et al., 2013; Myers, 2009; Ferber et al., 2009). The concept of global functional training (GFT) is geared to address biotensegrity in the body's myofascia architecture. Since the myofascial net infiltrates the entire body, and its elastic and sensorimotor properties are crucial to running, it makes sense that the GTF method of training is an effective and efficient approach (Kram and Dawson, 1998; Kubo et al., 2003; Schleip, 2003; Kjaer et al., 2009).

What do we mean by functional training?

Functional training is exercise that is specific to the body movement you are attempting to execute.

In more detail, functional training describes the concept of using multi-joint exercise (i.e. sports-specific exercises) which more closely reproduces the movement pattern of a sport (in our case running) and can be modulated to improve the sum of parts, all of the biomechanical movement pattern or physiological profile of the sport. For example, in running, the biomechanical pattern would be stride distance or frequency and the physiological profile would be aerobic power for a distance runner. In functional training, exercise must be global (i.e. using

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the whole body as much as possible) and not addressing isolated body regions.

In summary, functional training could be any sport-specific activity that moves an injured, deconditioned athlete or physically dysfunctioning individual towards safe return to sport or activities as soon as feasible.

What is the purpose of strength training for runners?

Running performance is dependent not only on a combination of aerobic and anaerobic capabilities, which vary based on the distance of the event, but also on other factors related to lower and upper body power and strength (Hudgens et al., 1987), speed and coordination.

It has been documented that force and power are strongly correlated with running performance for short distances (i.e. sprints, hurdles) (Meylan and Malatesta, 2008; Kale et al., 2009; Mikkola et al., 2011). For example, plyometric (Meylan and Malatesta, 2008) resistance (Mikkola et al., 2011) and explosive strength training (Spurrs et al., 2003; Buchheit et al., 2010) have shown significant improvements in sprint training performance (sprinting is not just about speed but strength, endurance, balance, etc.).

From observation, I firmly believe that strength training can help improve trunk pelvic complex, hip and lower extremity strength both concentrically and eccentrically, enhancing the structural integrity of the body's biotensegral architecture, thus improving running efficiency and performance.

The mechanism for this improved performance in distance runners is thought to be related to improved muscle (Dumke et al., 2010; Fletcher et al., 2010) and tendon stiffness (Dumke et al., 2010) and the elastic properties of the fascial net (Schleip, 2003; Huijing and Langevin,

2009; Tapale et al., 2010; Chapter 7). Therefore, the evidence for adapting exercises that train the fascial tissue net, as well as muscles and tendons, is paramount in decreasing and preventing running injuries and improving running economy. It should be noted that speed of running is a function of strength and coordination.

Sport-specific strength training for running must take into consideration sensorimotor factor and movement pattern behaviors (i.e. using global approach to exercise design) to guarantee the most transferrable effect of the training (Franklin and Wolpert, 2011; Smirniotou et al., 2008).

By contrast, middle and long distance running have received (have undergone or conducted) few studies that suggest force and power improve performance. However, a few well-designed studies have recently revealed that explosive strength training can improve the running economy of middle and long distance runners to a significant degree (Mikkola et al., 2007; Kelly et al., 2009; Ferraut et al., 2010) and also strengthen elastic elements like fascia, tendons and ligaments thus making more robust to withstand repetitive loading and injury (Arampatazis et al., 2007; Arampatazis et al., 2010; Hrysomallis, 2012; Kubo K et al., 2007).

What do we mean by myofascial training?

In the past, training for athletes focused mainly on conventional cardiovascular fitness, muscular strength, power and neuromuscular coordination. The classical biomechanical tradition of considering the body as functioning in separate segments, with attached levers, and concepts of linear mechanics is no longer feasible in the light of new research on myofascial function in the whole body (Schleip, 2003; Desmouliere et al., 2005; Huijing, 2007; Kubo et al., 2006). Humans

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like other species are complex biological and biotensegral systems (Levin, 2002). You cannot train body parts in isolation and expect to have efficient global functioning.

Running, in its true form, is mostly an elastic event (Bosch and Klomp, 2001; Legramandi et al., 2013). The mechanism of running involves the storing of energy during the deceleration or breaking phase (during foot ground contact) and instantaneous releasing energy during the lift-off phase initiated by ground reaction force (GRF) (Figure 31.1) (Legramandi et al., 2013). Using an elastic recoil technique (Chapter 7) will allow the runner to be more efficient, placing less stress on the musculoskeletal system and eventually decreasing injury risk. Training for runners must be elastically functional and global in its approach, inclusive of the entire body, and not just the lower extremities or individual muscle groups. It should be considered as complex training targeting neural adaptation, coordination, strength and proprioception modes.

There is evidence to support that different myofascial elements are affected by different loading styles and that fascia has an important role in maintaining muscle function (Steven et al., 1981; Ingber, 2008). Typical weight training loads the muscle in its normal range of motion, therefore strengthening fascial tissues arranged in series with active muscle fibers (Mackey et al., 2008; McBride et al., 2009). This type of loading has minimal effect on the intramuscular fibers that are arranged in parallel to active muscle fibers and also extra-muscular fascia (Huijing, 1999; Latrides et al., 2003; Fukashiro et al., 2006). This evidence reinforces that, during functional training of the runner's fascial net, exercises must have a dynamic varied loading pattern with rhythm to have an effect on the elastic components and resilience of the body's myofascial net.

The concepts of global functional training for the runner arise from these insights. The global functional training program, designed in this context, will address the running mechanism from a global perspective with exercises geared to train the runner's elastic myofascial component as well as the muscle, ligaments, bone and tendons. The functional myofascial exercise protocol for the runner is carried out with a certain amount of rhythm and an explosive component. Special attention is paid to the sport-specific movement pattern for running.

In maintaining form, or activated structural integrity, the body is able to set up its own internal and external anchor to create the dynamic stability needed as one segment generates power and the other stabilizes (Ingber, 2008; Kelsick 2006). This creates the alternate movement pattern biology designed for running. This patterning is classified as the concentric/eccentric; dynamic structural stability power/generating switching mechanism (between concentric and eccentric myofascial contraction) required for alternate body segmental movement in walking and running (Chapter 30). As previously described, running and walking are a simultaneous balancing dance between tension and compression in harmony with concentric and eccentric movement patterns of the entire body myofascial net and other supporting tissues. This tension and compression mechanism is always engaged during static or dynamic activity based on the principles of biotensegrity (Figure 31.1). In addition the body's tissues (which are soft matter) exhibit auxetic properties (i.e. they expand when tensioned (stretched) as opposed to shrinking-narrowing).

What global functional fascia training is specific to runners?

It has been documented that the manner, slow or fast, in which connective tissue is loaded

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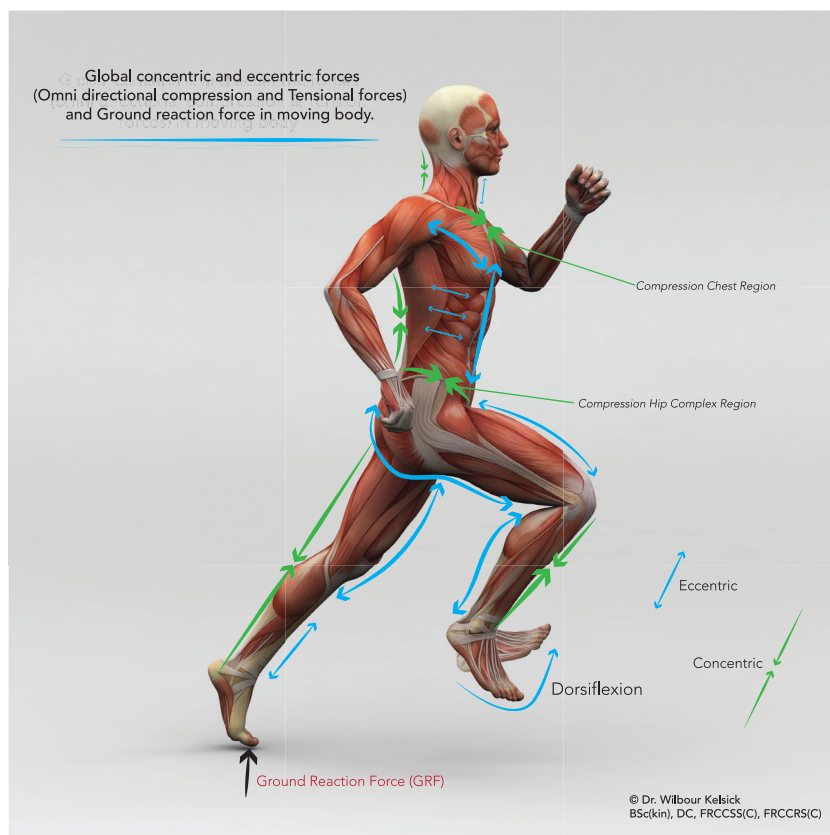


Figure 31.1 Some key global and compression omni directional external forces in a moving body in moving body

In terms of biotensegrity, a body in motion (walking, running, climbing, swimming, etc.) is a continuous well balanced harmonic dance between tension and compression of its global myofascial net and other supporting tissues.

will determine whether the tissue will become more elastic or react with hypertrophy (i.e. volume) (Kubo et al., 2003; Kjaer et al., 2009; Chapter 14).

In nature, kangaroos and gazelles are excellent examples of elastic storage and the release of energy during their movement patterns (Kram and Dawson, 1998; Chapter 8). The human myofascial net seems to have similar elastic behavior (i.e. kinetic energy storage and release) in our daily activities of walking, running or jumping (Chino et al., 2008; Sawicki et al., 2009). This justifies the approach of global functional training for the runner's myofascial net using explosive, rhythmic type exercise movement patterns (Fukunaga et al., 2002; Kawakami et al., 2002).

The functional fascia training concept

The elastic behavior of human fascia is now documented (Kubo et al., 2006; Chino et al., 2008). It stores energy and returns it quickly, as seen in cyclic movements like walking and running (Chapter 29; Kubo et al., 2006; Chino et al., 2008; Legramandi et al., 2013).

Running elastically uses less muscle power, that is, less metabolic energy (glucose) and more of the elastic fascia feature of the tissue, thus storing and returning energy back during propulsion. Global functional training for runners' fascia is designed to train the elastic myofascial net of the entire body (i.e. muscle, tendons, ligaments, bones). The modes of exercise include bouncing or plyometric movements, preparatory

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countermovement, unilateral movement patterns, coordination drills, and other exercises which mimic the mechanism of running (e.g. single leg hops, single leg squats, etc.). The exercise protocol avoids slow, jerky-type movement patterns, repetitive constant-angled movements, movement with mono tempo/rhythm, muscular dominant movement, segmental isolation-type movement pattern and minimizes constant loading thus encouraging variable loading of the runner's body. This is in line with research suggesting that the fascial system is better trained by use of a variety of vectors/angles, loads and rhythm (Huijing, 2007; Chapter 24).

We cannot delve into this topic without mentioning the importance that running, coordination posture and technique play in enhancing running efficiency and performance. In a correct sprinter's posture, for example, the athlete is at a considerable height off the ground in the flight phase, with a well-positioned body preparing for the landing phase (a very important body position to maximize horizontal distance travel through the air). Although it is not possible to discuss the concept of running technique in this context in this chapter, it should be noted that it is a crucial piece of the puzzle in preventing running injuries and improving performance.

Exercise protocol for training the runner's myofascia net

Sports conditioning programs need to be optimally individualized and specific because the limits of peak performance are highly variable even within the same discipline of sport. As described above, the purpose of appropriate exercises is geared to enhance sensorimotor coordination and to strengthen the elastic components of the runner's myofascial net. Energy is stored in the eccentric phase of motion and immediately released on the concentric phase.

The exercises are preceded by an eccentric pre-stretch (counter movement) (Chapter 24) that loads the muscle, tendon and fascia, preparing it for the ensuing concentric contraction. This coupling of the eccentric concentric muscle contraction is known as the stretch-shortening cycle, which physiologically involves the elastic properties of the connective tissues (fascia, tendon) and proprioceptive reflexes (Radcliffe and Farentinos, 1985; Tippet and Voight, 1995). The fact that connective tissue has a high capacity of adaptability and resilience makes it ideal for this type of training where loading forces, shearing and strain are highly variable. Connective tissue has the ability to continuously remodel its fibrous network when specific functional strain or load is applied to it (Chen et al., 1997; Langevin et al., 2010; Chapter 4).

In designing any exercise strengthening program, there are some basic exercise prescription guidelines which must be taken into consideration and a few important questions that need to be asked: Why are you doing the activity? What is the goal of the activity? Is it for fitness maintenance or for competition? Note that the basic principles of training will also apply here (i.e. principles of adaptation [acute and chronic], specificity, overload, and progressive overload, stress-rest, contraction, control, coordination, ceiling, maintenance, symmetry and overtraining) (Kraemer, 1994; Fleck and Kraemer, 1997). In this context we cannot address all these principles but they are found in detail in physiological texts on training (Fleck and Kraemer, 1997).

First, exercise prescription must consider the total demands of the program and ensure that the volume of exercise is not excessive, which can negatively interfere with the optimal physiological adaptation and performance. It should also consider the activity movement pattern, since complex systems like our bodies do not move in a linear manner or pattern.

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To ensure an effective prescription (Kraemer and Koziris, 1992; Tippet and Voight, 1995; Fleck and Kraemer, 1997), the following should be taken into consideration:

- the concept of periodization of the training program and goals of training
- developing a well-planned exercise recovery and rest protocol by using the principles of periodization
- understanding the balance between strength/power (intensity), coordination and aerobic and anaerobic (volume) training.

In addition, the key resistance training program components should be considered when designing functional fascia-type exercises (Kraemer, 1994; Fleck and Kraemer, 1997). These components are:

1. **Needs analysis:** this addresses questions about the myofascial net, whole-body segments to be trained, the energy/metabolic systems involved (aerobic, anaerobic), the type of muscle action (eccentric or isometric). Also the principle that the body is a complex biological system and is non-linear.
2. **Acute program variables:** this deals with choice, order, number of sets, rest period between sets and amount of load (intensity).
3. **Chronic program manipulation:** this addresses the principles of periodization as a means of designing long-term programs.
4. **Administrative concerns:** this deals with equipment needs in the gym (free weights, machine-assisted resistance isokinetics, jump platforms, etc.).

Exercise posture

These exercises are geared more towards middle distance and recreational runners, taking into consideration the concepts of transfer training

and exercise specificity. The concept of practicing one movement pattern to improve the efficiency of another movement pattern is known as “transfer of training”. For example, a runner will get little benefit from doing high-intensity seated rowing. Split squats or walking lunges, which are more similar to the movement pattern of running, would be more beneficial. When exercise movement patterns are similar, this principle is known as “exercise specificity”. Specificity in athletic or any type of training is paramount and a main guarantee that one can achieve transfer of training. (Franklin and Wolpert, 2011). Due to space constraints in this chapter, only eight exercises have been documented.

All exercises are performed in a “closed kinetic chain” posture to increase joint compressive forces, improve joint congruency and myofascial co-contraction/activation, which will overall enhance the dynamic stability of the body segments targeted (modified from Lefever, 2011) although the entire body is involved (the global approach). The concept of transfer of training and exercise specificity is taken into consideration. Proper foot placement is paramount. The ankle should be in a “locked” position attained by dorsi-flexion. This allows a stiff but dorsiflexed forefoot to contact the ground, instantaneously transmitting the GRF up the kinetic chain. This well-timed, coordinated action of the foot creates the stiffness in the lower extremity chain required to create reactive strength to receive the GRF (Winkler, unpublished data). Such action ensures a pre-stretch and engages the elastic components of calf. Engagement/activation of the neuromuscular components of the trunk, pelvis, pelvic floor and lower extremities is also necessary. The exercises are not segmental but address the entire body running movement pattern.

Exercise mode

All exercises are performed explosively with quick repeated rhythm/tempo to enhance the elastic effect on the muscular and fascial tissues.

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Pre-exercise preparation

This preparation should be about 15 min and consists of the following to enhance general body mobility, ankle and foot ground reactivity:

- 3–5 min jogging
- Upper body alternating arm action (simulating arms during running) with rhythmic breathing
- Low amplitude ankle/foot bouncing with initial active dorsiflexion and plantar flexion when landing on ground
- Standing lower extremity swings from hips
- Walking lunges with quick tempo
- Sideway scissors runs – lower extremity crossovers
- Double and single ankle hops with dorsiflexion of foot landing on balls of the feet
- Hip mobility drills – flexion and extension hip swings.

Functional training exercises

Some of these exercises were developed in collaboration with the coach, Gary Winckler. These exercises are designed to target and exploit the elastic components in the fascia net, muscle and tendon tissues. Execution must be explosive, rhythmic and reactive. Attention to global body posture and technique execution is crucial to achieve the full benefits.

1. Extended arm overhead with resistance: rectus abdominis and anterior compartment eccentric strengthening (Figure 31.2)

Purpose

To build anterior trunk anchoring eccentric strength for abdominal and lateral wall muscles (transversus, internal and external obliques), which transmit forces from the back structures



Figure 31.2

Over-head pull: eccentric resistance for anterior abdominals/trunk.

(mainly the thoraco-lumbar fascia). The rectus abdominis, pelvis and hip complex all work eccentrically to improve balance, coordination and force transmission during the single leg stance in the walking movement of this exercise.

Position/posture

Stand upright, engage trunk anterior and posterior musculature, pelvic floor and hip structures. Hands are overhead with extended elbows.

Start position

Stand facing away from anchor of elastic tubing or pulley weight cables, which is anchored at the height of extended forearm overhead. Alternatively, you can use an individual assistant (as an anchor) to hold the tubing.

Movement technique

- Flex one hip with flexed knees (e.g. like running stance) with your thigh in midline and your pelvis level.
- Check your alignment then begin to march or walk against the overhead resistance.

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- Lift the chest and chin up and look slightly upward so you feel the tension in the anterior abdominal wall complex.
- Increase pace from a walk to a slow jog maintaining tension on the elastic strap at all times.

The key is to be as upright as possible and to try not and over-extend the spine/trunk but maintain good tension of the elastic strap as you move forward. You can also use cable pulley weights for this exercise.

Dosage

Set: 3–6

Reps: 10–20 steps

Rest: 1–2 min

2. Quick step-ups

Purpose

This is a close kinetic-chain movement to strengthen trunk, pelvis, hip, knee, ankle and foot, and to improve coordination and balance in single leg stance.

Position/posture

Stand upright and engage anterior and posterior trunk musculature, pelvic floor, hip structures.

Start position

Stand facing a step or box.

Movement technique

- Your front leg on the step/box (12–14") with your thigh in midline and your pelvis level back leg on ground.
- Check your alignment then quickly push with the back leg, lifting it off the ground and pushing your body straight upward.
- Step off with the leg, which was first on the box.
- Repeat alternating your legs.
- The key is to push off with the back leg/leg on the ground and not the front leg/leg on the box.

Dosage

Set: 3–6

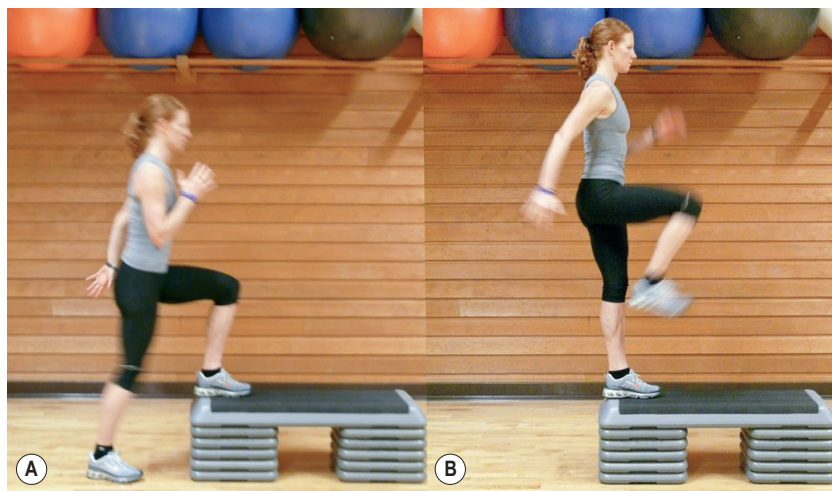


Figure 31.3

(A) Quick step-ups and (B) walking step-ups.

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Reps: 10–20

Rest: 1–2 min.

3. Low amplitude double leg hop over mini-hurdles

Purpose

A plyometric drill, which initiates the stretch-shortening cycle of the lower extremity complex. This drill assists in developing elastic strength, speed and explosive power of the lower leg and the pelvis, especially the gluteals, hamstrings, quadriceps and gastrocnemius-ankle complex. Enhances the elastic fascial components in the trunk, pelvis, hip, knee, ankle and foot.

Position/posture

Stand upright and engage the trunk's anterior and posterior musculature, pelvic floor, hip structures. Step up two mini-hurdles 6–12 inches high about one stride length apart.

Start position

Stand upright about half a stride length in front of the hurdles with shoulders slightly forward,

head up. Elbows should be at 90° and hands at your sides with thumbs up.

Movement technique

- Begin by performing a counter-movement downward and jump as high as possible, flexing legs so the feet arrive under the buttocks. Bring the knees up medium high and forward for each jump to ensure maximum lift.
- To land, ensure the ankle is dorsiflexed. Jump forward again with the same cycle of leg and foot pattern.
- Execute as rapidly as possible, always moving forward.
- The key is to gain moderate height and maximum distance without affecting repetition rate.

Dosage

Set: 3–6

Reps: 10–20

Rest: 1–2 min.

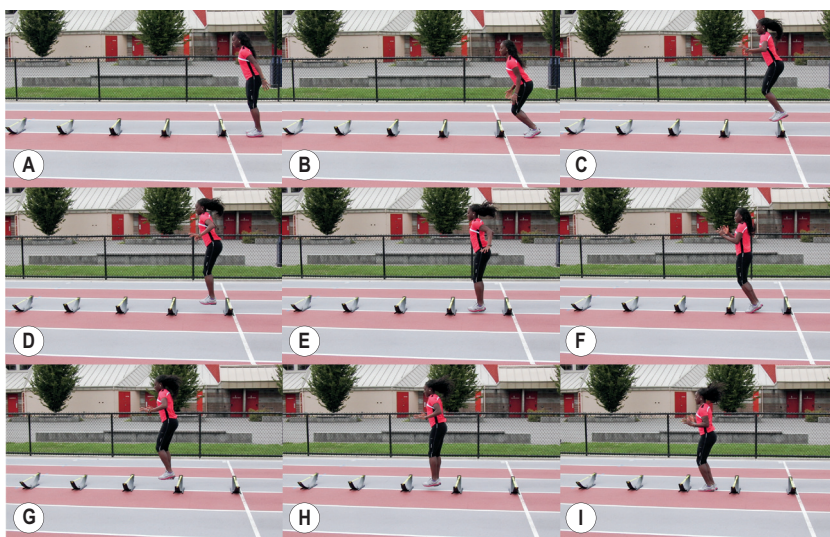


Figure 31.4

(A–I) Double leg hops over mini-hurdles.

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4. Double leg ankle hops (Figure 31.5)

Purpose

A plyometric drill, which initiates the stretch-shortening cycle of the lower extremity complex. It helps develop elastic strength, speed and explosive power of the lower leg and gastrocnemius-ankle complex. It enhances the elastic myofascial components of the lower knee, ankle and foot.

Position/posture

Stand upright and engage trunk anterior and posterior musculature, pelvic floor, hip structures.

Start position

Stand upright with both feet on the ground and hands by your side.

Movement technique

Push off the ground and immediately dorsiflex the ankle joint. The knee should be in extension. On landing, ensure your foot is dorsiflexed and land on the balls of your feet. Repeat the sequence rapidly, maintaining an extended knee. Remain basically in the same spot.

Dosage

Sets: 4

Reps: 15–30

Rest: 2 min.

5. Single leg ankle hops (Figure 31.6)

Purpose

A plyometric drill, which initiates the stretch-shortening cycle of the lower extremity

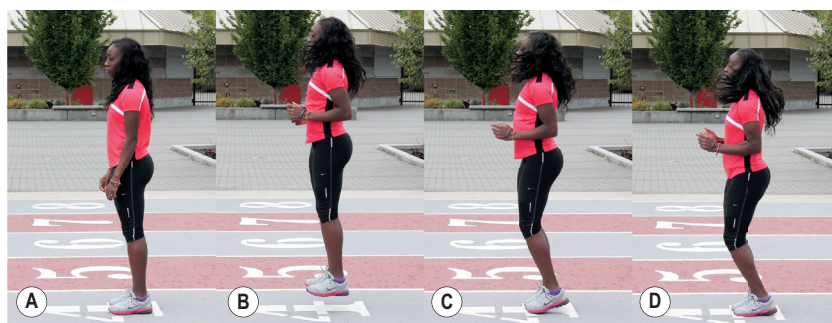


Figure 31.5

(A–D) Double leg ankle hops.

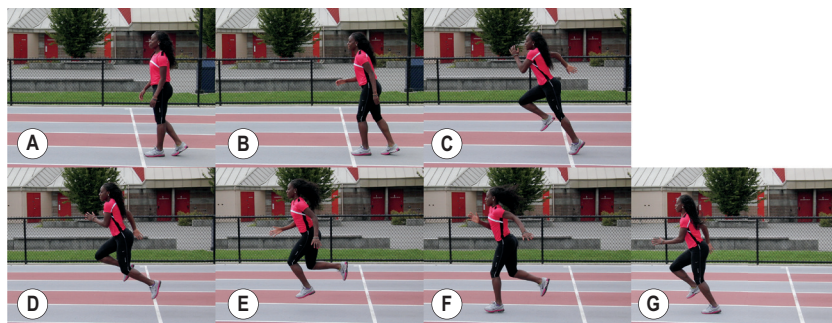


Figure 31.6

(A–G) Single leg ankle hops.

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complex. It helps develop elastic strength, speed and power of the lower leg and gastrocnemius-ankle complex. It enhances the elastic fascial components of the lower knee, ankle and foot.

Position/posture

Stand upright and engage the trunk's anterior and posterior musculature, pelvic floor, hip structures.

Start position

Stand upright with both feet on the ground and hands by your side.

Movement technique

Push off the ground on one leg, only leaping forward and immediately dorsiflexing the ankle joint. Try to land one stride length ahead on the ball of your foot. The knee should be in extension. On landing, ensure your foot is dorsiflexed and land on the push-off leg on the ball of your feet. Repeat the sequence rapidly, maintaining an extended knee and alternating legs.

Dosage

Sets: 4

Reps: 15–30

Rest: 2 min.

6. Double leg jumping split lunges on the spot (Figure 31.7)

Purpose

A plyometric drill, which initiates the stretch-shortening cycle of the hip and lower extremity complexes. This drill helps to develop elastic strength, speed and power of the lower leg and pelvis, especially the hip flexors, gluteals, hamstrings, quadriceps and gastrocnemius-ankle

complex. It enhances the elastic strength of fascial components in trunk, pelvis, hip, knee, ankle and foot complexes. The goal is to attain maximum height.

Position/posture

Stand upright and engage trunk anterior and posterior musculature, pelvic floor, hip structures, as in the previous exercises.

Start position

From a standing parallel foot position and shoulder width apart get into lunge posture. Step forward with say the **right** leg knee at 45–90° at the hip and 45–90° and hip flex about the same degree and your **left** back hip is now extended (to a comfortable near maximum range of motion) and knee of back **left** leg also flex at 45–90°. In this position you are maintaining an upright/ erect and engaged trunk and pelvis. Your arm positioning is important. The arm (left) on the opposite side of the leg that's in the back (extended hip) should be flexed 90° at shoulder and elbow. The arm (right) on the same side of the leg that's in front (flexed hip) should be extended about 50–60° and 90 degrees flexion of the elbow (basically your posture should be the running stance).

Movement technique

The goal is to jump into the next lunge position landing on the same area/spot by pushing off the back leg explosively lifting your body into the air and again landing in a lunge position. Using counter-movement technique, drop down into a lunge position and stop that movement subsequently exploding upward as far as you can with a scissor-like motion. Note while in the air you need to execute a scissor action. Quickly bring the **right** hip which was in flexion in extension

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Figure 31.7

(A–L) Double leg jumping split lunges on the spot

and the extended **left** hip into flexion before landing again in a lunge posture but this time your **left** hip and leg is now in front (flexed **left** hip) and your **right** hip and leg are now in the back (extended **right** hip). This scissors movement (alternating leg position front and back) is repeated continuously for desired number of reps. It is important during landing (should be on ball of feet) and pushing off that the ankle complex is held in dorsiflexion posture as much as possible. This exercise can be performed by moving the forward covering distance but is much more demanding (as for quick walking lunge) for desired number of repetitions.

Dosage

Sets: 4

Reps: 5–1

Rest: 2 min.

7. Quick walking lunge (Figure 31.8)

Purpose

This is a close kinetic-chain movement in lower extremity complex. It helps to develop the elastic strength, speed and power of the lower leg and pelvis, especially the hip flexors, gluteals,

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hamstrings, quadriceps and gastrocnemius-ankle complex. It enhances the elastic strength of myofascial components in trunk, pelvis, hip, knee, ankle and foot. The goal is to attain a good rhythm.

Position/posture

Stand upright and engage the trunk's anterior and posterior musculature, pelvic floor, hip structures as in the previous exercises.

Start position

Place feet shoulder width apart, bend one leg to 90° at the hip and 90° at the knee, attaining more or less a running stance.

Movement technique

Lunge forward quickly with the unsupported leg. As soon as contact is made with the ground, recover the hind leg and use it to repeat the lunge.

Land with feet/ankle in dorsiflexion so you land on the balls of the feet in a split lunge position and immediately repeat the sequence, initiating the pushing phase on the front leg to propel the body forward. Cover about 30–40 m.

Dosage

Sets: 4

Reps: 15–30 (30–40m)

Rest: 2 min.

8. Alternate leg box jumps (Figure 31.9)

Purpose

A plyometric drill, which initiates the stretch-shortening cycle of the lower extremity complex. This drill helps to develop the elastic strength, speed and explosive power of the lower leg and

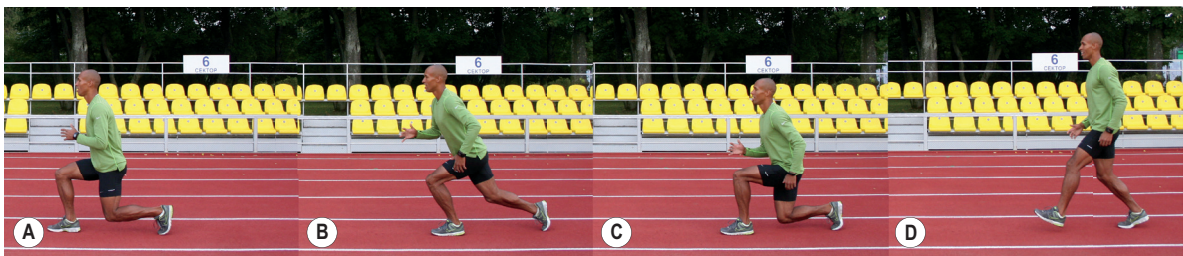


Figure 31.8

(A–D) Quick walking lunge.

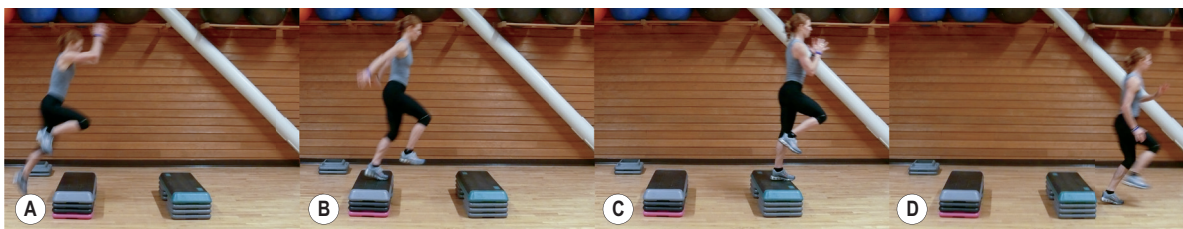


Figure 31.9

(A–D) Alternate leg box jumps.

Functional training methods for the runner's myofascial systems

pelvis, especially the hip flexors, gluteals, hamstrings, quadriceps and gastrocnemius-ankle complex. It enhances the elastic fascial components in the trunk, pelvis, hip, knee, ankle and foot.

Position/posture

Stand upright and engage trunk anterior and posture musculature, pelvic floor and hip structures.

Start position

Stand upright on one leg, with one leg in front of the other, as if taking a step. Shoulders are oriented slightly forward and head faces upwards. Arms are at the sides.

Movement technique

Begin the exercise by pushing off with the back leg. Drive the knee up to the chest to achieve maximum height and distance before landing. Quickly extend the driving foot outward. Cycle the arms in contra-lateral motion in the air for balance. Repeat the sequence using alternate legs on landing.

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Dosage

Sets: 2–4

Reps: 8–12 (40 m)

Rest: 2 min.

Summary

Speed is a composite of strength and coordination. Walking, running and sprinting are movements in complex biological systems where the principles of biotensegrity (internal omni-directional forces, tension and continuous compression) function in a harmonic dance of balance, to maintain the integrity of the body's architectural systems during motion. In addition these internal forces prepare the body's closed kinetic chain system to absorb, transmit and create movement from the external GRF. Effective training for runners must address complex global movement pattern similar to running and not individual muscle groups. Training the myofascia net re-enforces strength and enhances elasticity of the biotensegral architecture of the running body systems. This, in turn, results in improved running economy and efficiency and decreases running injuries.

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